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DEACTIVATION OF RIPENED UTERINE TISSUE BY CALCIUM CHANNEL BLOCKER VERAPAMIL

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ABSTRACT:

Calcium has an integral role in the contraction of smooth muscle in general, and ripened uterine muscle in particular. The objective of this study was to evaluate the role of calcium channel blocker "Verapamil" in the isolated uterine tissue of rat after stimulation by oxytocin. Tissuse were obtained from albino rats of Sprague Dawley strains. The activity was recorded on polygraph showing an enormous increase in the activity when tissue was challenged with oxytocin and reduced this activity when calcium channel blocker verapamil was added It is concluded that the calcium channel blocker verapamil can be useful in suppressing the premature increased activity in preterm labor.

INTRODUCTION

Preterm labor continues to complicate a significant number of pregnancies, this leads to premature birth, which accounts for a large fraction of perinatal morbidity and mortality (Abel M.H. and Hollingsworth 1985). Among the various causes of preterm labor, one of the factors is excessive muscle contractility. In uterine smooth muscle contractile protein calmodulin reacts with calcium ions. This complex initiates an interaction by activating myosin kinases (a phosphorylating enzyme) .In response to this enzyme, myosin head becomes phosphorylated which binds with actin filament and muscle contraction occurs (Bolton, TB, 1979). When calcium concentration falls below a critical level the process of contraction reverses back and contraction ceases (Kamm and Stull 1985).

Mechanical activity in uterine smooth muscles is dependent on the extracelluler calcium entering the cytoplasm through calcium channels (Dawood *et al.,* 1979). The availability of extracelluler calcium or presence of blockers of calcium channel strongly influences the response of uterine smooth muscles (Husszar and Roberts 1982%

Sarcoplasmic reticulum occupies 1.57.5% of total volume of smooth muscle and releases the calcium with the onset of impulse. This is particularly abundant in pregnant or estrogen primed uterine muscle fibers. This system stores the calcium by sequestration during relaxation. The increase in free calcium is a prerequisite for uterine contraction. Prostaglandin and oxytocin are two physiological hormones that cause uterine contraction. Oxytocin concentration rise in maternal plasma with gestation. Oxytocin acts through specific plasma membrane receptors to stimulate phosophatidyl inositol which gets hydrolyzed and thereby causes formation of inositol phosphate. This in turn causes release of ca++ from intercellular stores. Oxytocin also acts to inhibit the plasma membrane ca+ + adenosine triphosphatase (calcium ATPase) pump which leads to increase intracellular calcium level. Increased sensitivity of the myometrium to oxytocin occurs with advancing pregnancy Women who deliver prematurely have highest sensitivity to oxytocin (Fuchs *et al.,* 1983).

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The prostaglandin E2(PGE2), F2 and oxytocin inhibit calcium uptake and enhance calcium release . Calcium channel blockers can be tried to inhibit uterine muscle contraction in preterm labor. These potential tocolytic agents have been shown to suppress both prostaglandin and oxytocin induced uterine activity in isolated human and inhibit myometrial preparation (Granger *et al.,* 1985).

PURPOSE OF STUDY

This is to evaluate the efficacy of a drug verapamil a calcium channel blocker in blunting the oxytocin induced contraction in ripened uterine muscle.

METHODOLOGY

This was carried out on isolated uterine tissue of rat. Full term between day 22-25 female albino rats of sprague dawley strain were used. The animal house of the Jinnah postgraduate medical center Karachi, supplied the animals.

The animals were killed by blow on the head Abdomen was opened and fetus separated. The whole uterus was transferred to oxygenated De-Jalons solution, 25 mm strip was taken from the middle portion of the uterus and mounted in the organ bath connected to the polygraph through a transducer.

Temperature of the organ bath and pH of solution was maintained at 37°C and 7.4 respectively. The tissue was allowed to equilibrate in the organ bath for half an hour. The activity was recorded by grass polygraph 7B-USA. First normal contractile response of the isolated ripened uterine tissue was recorded for ten minutes. Then oxytocin was added to organ bath and its effect recorded for another ten minutes. After this, verapamil was added to the same above preparation and response was recorded for the subsequent ten minutes.

The effect of all the drugs were observed on the rate and amplitude of the ripened uterine muscle contraction.

RESULTS

In control group after taking normal activity tissue was exposed to oxytocin. Eight experiments were performed. The activity was recorded for ten minutes. The mean value of spontaneous rate of contraction was 8.00 ± 0.25. And after exposure to oxytocin 0.1 i.u/ml this increased to 11.75 ± 0.23. This difference is statistically highly significant (P<0. 001). In case of amplitude of contraction the mean value of control was 12.96 ± 0.44 mm. After exposure to oxytocin 0.1 iu/ml. height of contraction rose to mean value of 20.07 ± 0.32. This difference is also highly significant (P<0.001).

Rate of spontaneous contractions/ 10minutes were recorded with verapamil after exposure to oxytocin. The rate was reduced from a mean value of 11.7 ± 0.23 with oxytocin to the mean value of 5.20 ± 0.18 with verapamil, where as the mean value of control was 8 ± 0.25. This reduction in the rate with verapamil after exposure to oxytocin was highly significant (P<0. 001), table and fig a.

The amplitude of contraction also showed a significant reduction by Verapamil. The amplitude reduced to a mean value of 6.26 ± 0.44 m. m as compared to oxytocin which was 20.07 ± 0. 32m.m. In these experiments activity in the control gave a mean value of 12.96 ± 0.44 m. m. This difference is also highly significant (P < 0. 001), table and figure b.

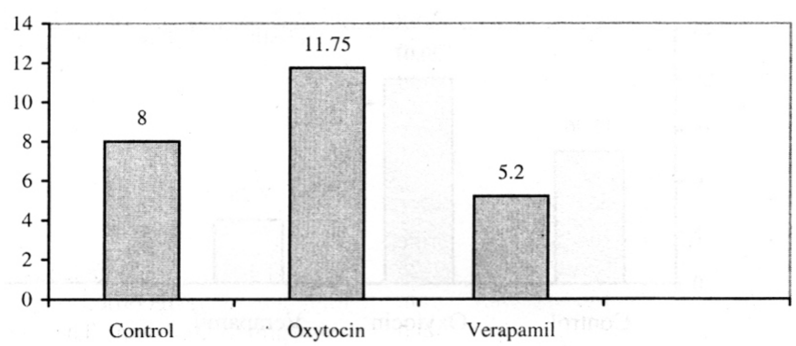
DISCUSSION

Infants born prior to term accounts for a majority of neonatal deaths and a large percentage of preterm infants suffer significant morbidity and long term sequelae such as motor and intellectual handicaps (Lee Ktseng *et al,* 1976). This incidence can be reduced by adequate prenatal care with active treatment of

Table (a)
Effects of verapamil 0.25 hg/ml on rate of contraction on isolated ripened uterine muscle of rat after exposure to oxytocin 0.1 i.u/ml

No. of Contractions/B Minutes			
Control	Oxytocin	Verapamll	P Value
8.00 ± 0.25	11.75 ± 0.23	5.20 ± 0.18	<0.001
(8)	(8)	(8)	

Contractions/ 10mins



factors that predispose to preterm labor, one of the factors being excessive uterine contractility (Steve N Caritis 1979).

An increase in the number of myometrial oxytocin receptors was reported in women who had delivered preterm, indicating the increase in uterine activity (Takahashi et al., 1980). This correlates with our observation in which when tissue from ripened uterus was challenged with oxytocin, it showed an increase in the rate as well as in amplitude of contraction. This indicates the presence of oxytocin receptors and their enhanced sensitivity to oxytocin.

We observed that verapamil was very efficacious in decreasing the rate and amplitude of contraction, when added to ripened uterine tissue already exposed to oxytocin. This shows that extracelluler calcium has a very powerful role in the contraction of uterine muscle and certain concentration of calcium is necessary for uterine contraction. This concentration decrease with the calcium channel blocker verapamil. This also shows that the intracellular calcium is not sufficient for contraction, which is released by oxytocin. Our observation regarding the relaxing efficacy of verapamil on isolated uterine tissue is in consensus with the studies carried out by different researchers (Abel M.H., Hollingsworth 1985, Cspao A.I. *et al.,* 1982 and Granger *et al.,* 1982).

This shows that the drug may have some role in labor and it could be implicated to prevent preterm labor and preterm births in humans. Studies can be carried out on humans to prove the safety of the drug in a dose necessary for tocolysis.

Table (b)
Effects of verapamil 0.25 ug/ml on amplitude of contraction on isolated ripened uterine muscle of rat after exposure to oxytocin 1 i.u/ml

Amplitude of Contractions Inmm.			
Control	Oxytocin	Verapamil	P Value
12.96 ± 0.44	20.07 ± 0.32	6.26 ± 0.41	< 0.001
(8)	(8)	(8)	

Amplitude (mm)

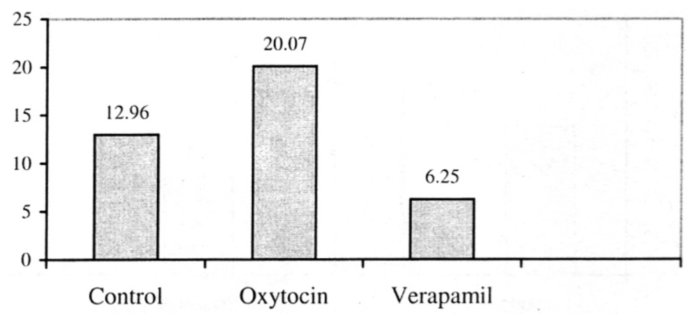


Fig. (b): Effects of verapamil 0.25 ug/ml on rate of amplitude of contraction on isolated ripened uterine muscle of rat after exposure to oxytocin 0.1 i.u/m1

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